AI Assisted Coding

Name - Neerudu Gifty Taniksha
Roll Number - 2503A51107
Batch - 38

Question 1

```
# This program calculates the total marks of a student in 5
subjects.

subject1_marks = float(input("Enter marks for Subject 1:
"))
subject2_marks = float(input("Enter marks for Subject 2:
"))
subject3_marks = float(input("Enter marks for Subject 3:
"))
subject4_marks = float(input("Enter marks for Subject 4:
"))
subject5_marks = float(input("Enter marks for Subject 5:
"))

total_marks = (
    subject1_marks
    + subject2_marks
    + subject3_marks
    + subject4_marks
    + subject5_marks
)

print("Total Marks:", total_marks)
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ c:/Users/tanik/.local/bin/python3.13.exe "c:/Users/tanik/OneDrive/Desktop/c_prog/Github/ai/task 1.py"
Enter marks for Subject 1: 70
Enter marks for Subject 2: 95
Enter marks for Subject 3: 97
Enter marks for Subject 4: 98
Enter marks for Subject 5: 95
Total Marks: 455.0
```

Question 2

```python
class BankAccount:
    def __init__(self, account_number, account_holder, bala
nce):
        self.account_number = account_number
        self.account_holder = account_holder
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
        print("Deposited:", amount)
        print("Current Balance:", self.balance)

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
            print("Withdrawn:", amount)
            print("Current Balance:", self.balance)
        else:
            print("Insufficient balance.")


account = BankAccount("123456", "Karthik", 5000)

account.deposit(1000)
account.withdraw(2000)
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ c:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/ai/task2.py
Deposited: 1000
Current Balance: 6000
Withdrawn: 2000
Current Balance: 4000
```

Question 3

```python
numbers = [10, -5, 0, 8, -2, 0]

positive_count = 0
negative_count = 0
zero_count = 0

for number in numbers:
    if number > 0:
        positive_count += 1
    elif number < 0:
        negative_count += 1
    else:
        zero_count += 1

print("Positive:", positive_count)
print("Negative:", negative_count)
print("Zero:", zero_count)
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ c:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/ai/task3.py
Positive: 2
Negative: 2
Zero: 2
```

Question 4

```python
def check_number(number):
    if number % 2 == 0:
        return "Even"
    else:
```

```
        return "Odd"


print(check_number(10))
print(check_number(7))
print(check_number(0))
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ c:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/ai/task4.py
Even
Odd
Even
```

Question 5

```
def calculate_discount(price):
    if price > 1000:
        discount = price * 10 / 100
        return price - discount
    else:
        return price


print("Price for ₹800:", calculate_discount(800))
print("Price for ₹1500:", calculate_discount(1500))
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/De  Open file in editor (ctrl + click)
$ c:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/ai/task5.py
Price for ₹800: 800
Price for ₹1500: 1350.0
```

Question 6

```
class Book:
    def __init__(self, title, author):
        self.title = title
        self.author = author
```

```
        self.available = True

    def issue_book(self):
        if self.available:
            self.available = False
            print(f'"{self.title}" has been issued.')
        else:
            print(f'"{self.title}" is already issued.')

    def display_book(self):
        status = "Available" if self.available else "Issue
d"
        print(f"Title: {self.title}")
        print(f"Author: {self.author}")
        print(f"Status: {status}")
        print()


books = [
    Book("Harry Potter", "J.K. Rowling"),
    Book("The Alchemist", "Paulo Coelho"),
    Book("1984", "George Orwell")
]

print("Library Books:")
print("----------------")

for book in books:
    book.display_book()

print("Issuing books:")
print("----------------")

books[0].issue_book()
books[0].issue_book()
books[1].issue_book()

print("\nUpdated Library:")
```

```python
print("----------------")

for book in books:
    book.display_book()
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ c:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/ai/task6.py
Library Books:
---------------
Title: Harry Potter
Author: J.K. Rowling
Status: Available

Title: The Alchemist
Author: Paulo Coelho
Status: Available

Title: 1984
Author: George Orwell
Status: Available

Issuing books:
---------------
"Harry Potter" has been issued.
"Harry Potter" is already issued.
"The Alchemist" has been issued.

Updated Library:
---------------
Title: Harry Potter
Author: J.K. Rowling
Status: Issued

Title: The Alchemist
Author: Paulo Coelho
Status: Issued

Title: 1984
Author: George Orwell
Status: Available
```